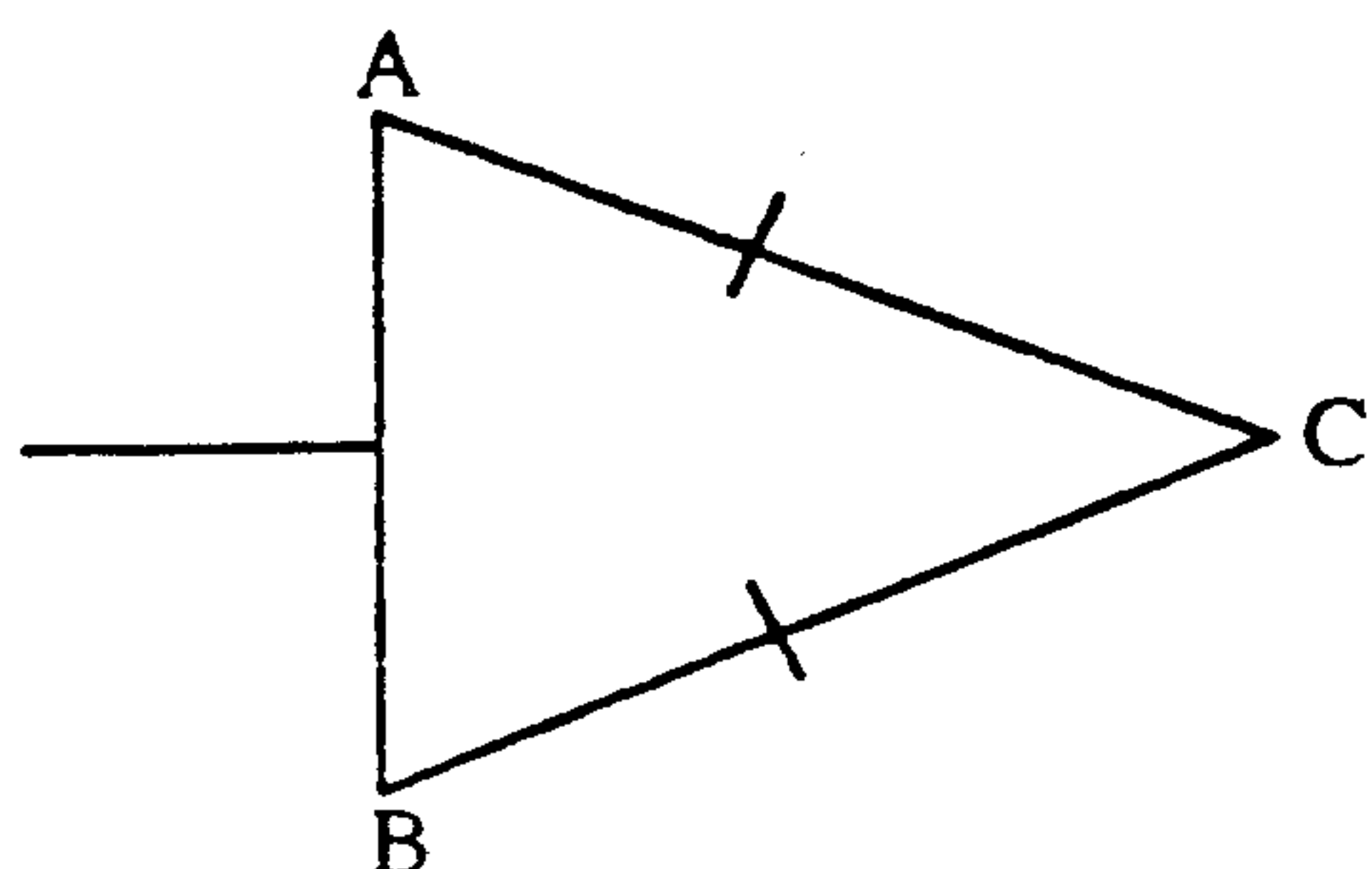


**Question 1.** ( Start a new page. )

- a) Given that P, Q, R are the points (1,3), (2,0), and (5,1) respectively :
- show that the line PQ is perpendicular to QR;
  - prove that the triangle PQR is isosceles;
  - find the coordinates of S, such that PQRS is a square.

- b) A triangle LMN is right angled at L.  
If  $LM = x$ ,  $NM = x + 3$ , and  $NL = 5$ , find  $x$ .

c)



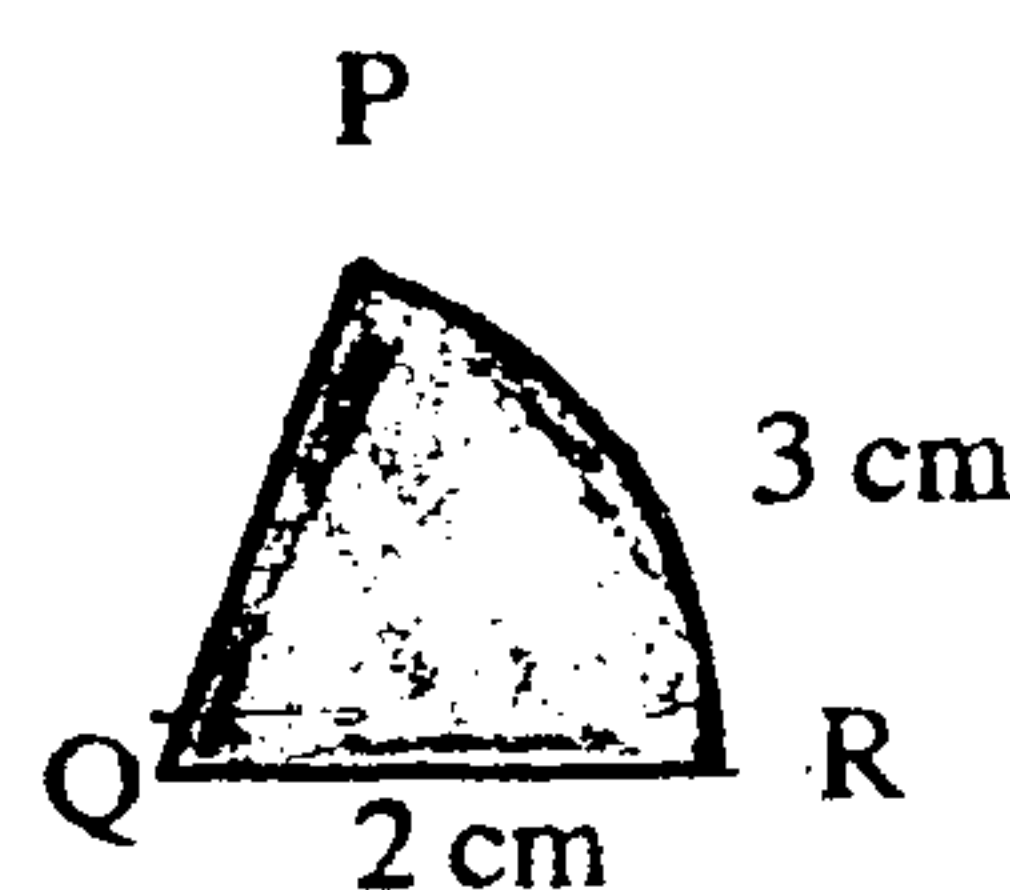
Find the exact area of the arrow head, where  $AC = BC = 6$  cm and  $\angle ACB = 45^\circ$ .

- d) Joan invested \$5,000 at the beginning of 1985 at 8% per annum, compounded at the end of each year.  
What will her investment be worth at the end of 1995 ?

**Question 2.** ( Start a new page. )

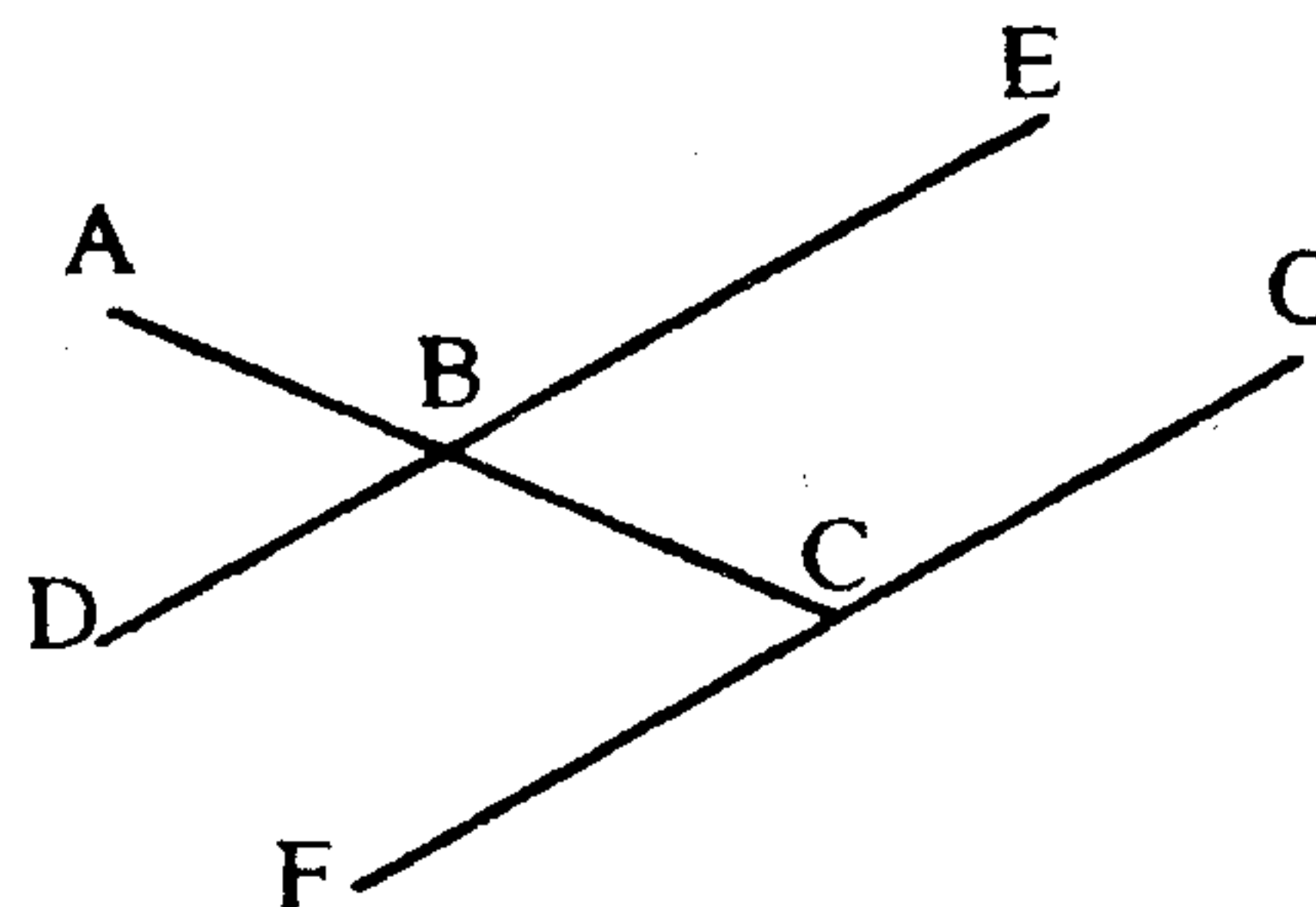
- a) Differentiate: (i)  $x^2 e^x$  (ii)  $\ln(\cos x)$

- b) The shaded figure, PQR is a sector of a circle, radius 2 cm.  
Arc PR is 3 cm in length.



- Find angle  $\hat{PQR}$  in radians.
- Find the area of the sector.

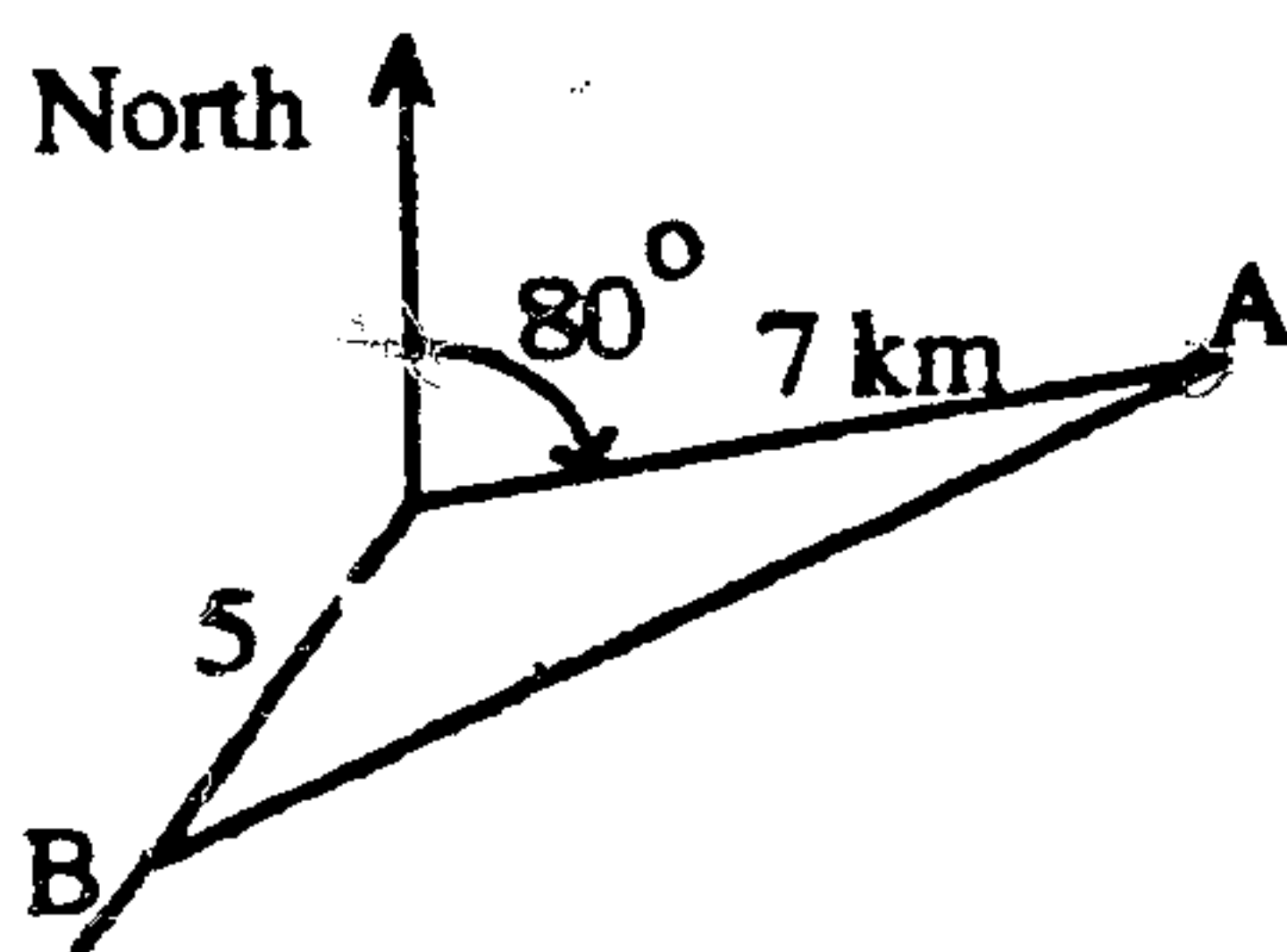
- c)  $\angle ABD = 47^\circ$  and  $\angle BCG = 133^\circ$   
Prove that lines DE and FG are parallel.



- d) Simplify (i)  $\sin(\pi - x)$  (ii)  $\ln e^{3x}$

**Question 3.** (Start a new page.)

- a) (i) Integrate  $(4x+3)^5$ .  
 (ii) Find the exact value of  $\int_0^{\pi/6} \cos 2x \, dx$ .
- b) A ship A is 7 km away from a lighthouse L on a bearing of  $080^\circ$  and a ship B is 5 km away from the lighthouse on a bearing  $210^\circ$  as shown in the diagram.

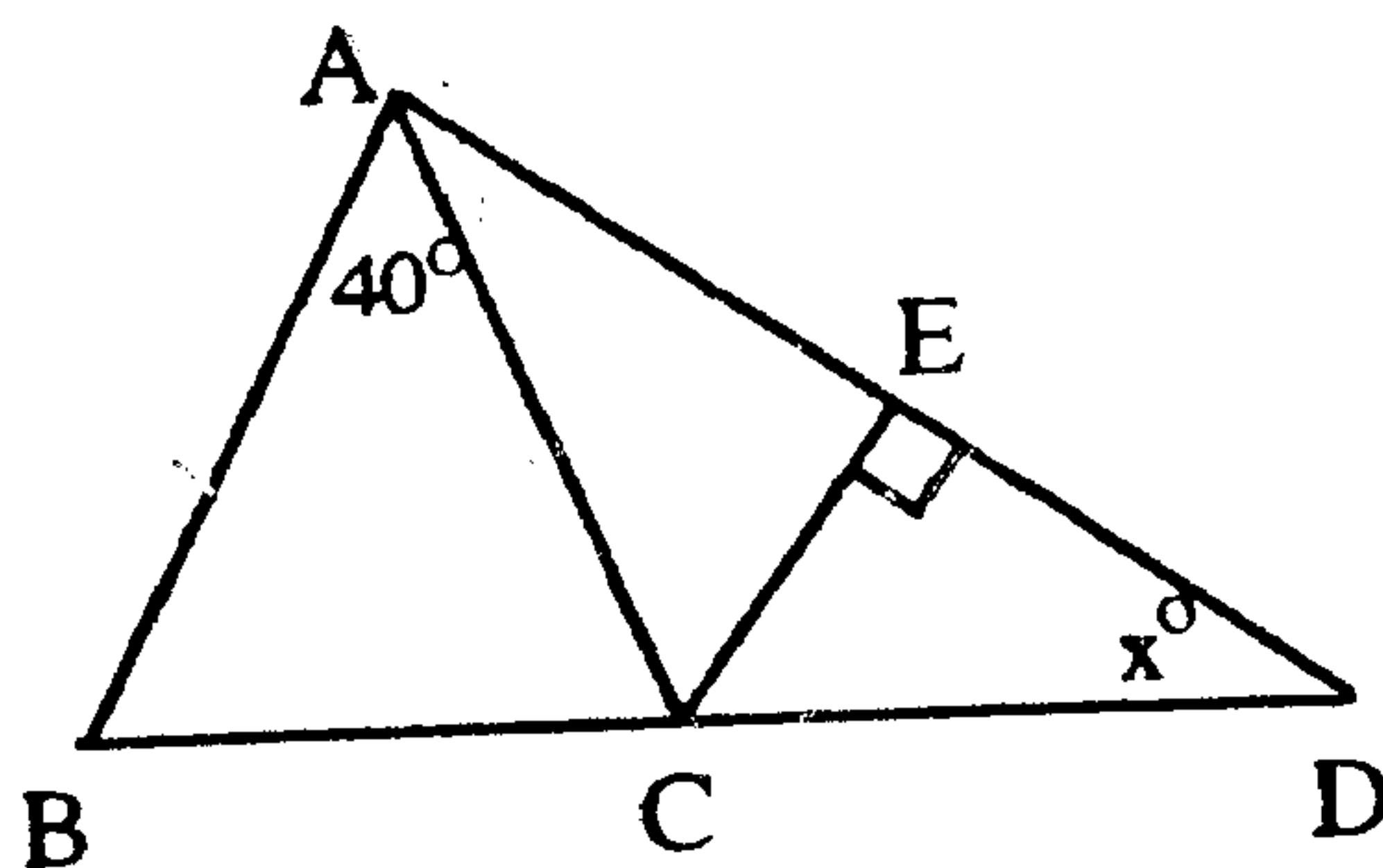


- (i) Find the distance between the ships A and B to the nearest km.  
 (ii) Find the bearing of the ship A from the ship B, to the nearest degree.
- c) Solve  $2\cos 2x = 1$  for  $0 \leq x \leq 180$ .

**Question 4** (Start a new page.)

a)

(not to scale)



$AB = AC$   
 $CE$  bisects  $\angle ACD$   
 $CE \perp AD$ .

- (i) Find  $x^\circ$ , giving reasons.  
 (ii) Prove that triangle ACD is isosceles.
- b) The gradient of a curve is given by  $\frac{dy}{dx} = 3x^2 - 4$ .
- (i)(a) Find  $\frac{d^2y}{dx^2}$   
 (b) Find the values of  $x$  for which the curve both increases and is concave downwards. Give reasons for your answer.
- (ii) If the curve passes through the point  $(1, -2)$ , find the equation of the curve.

**Question 5** ( Start a new page. )

- a) Consider the curve  $y = 4x^3 - 3x^4$ .
- (i) Find the points where the curve cuts the  $x$ -axis.
  - (ii) Find any stationary points and determine their nature.
  - (iii) Find all points of inflexion.
  - (iv) Sketch the curve showing the above results.
- b) Find the area defined by the curve  $y = 4x^3 - 3x^4$  and the  $x$ -axis.

**Question 6.** ( Start a new page. )

- a) To start a game a player has to throw a 6 with a die.  
Find the probability that the player starts at :
- (i) his first throw.
  - (ii) his second throw.
- b) The displacement of the piston shown, starting from the middle of the cylinder, is modelled by the equation  $x(t) = \frac{1}{4}\sin(\pi t)$ ,  
where  $x$  is in metres and  $t$  in seconds.



- (i) Sketch the displacement-time function for the first 2 seconds.
  - (ii) Calculate the total distance over which the piston moves in the first 5 seconds.
  - (iii) Calculate the maximum speed of at which the piston moves.
- c) A tree was 12 metres high at the start of a year and it increased by 1.5 metres during that year. If in each succeeding year the growth is  $\frac{3}{4}$  of that during the previous year, find the limiting height.

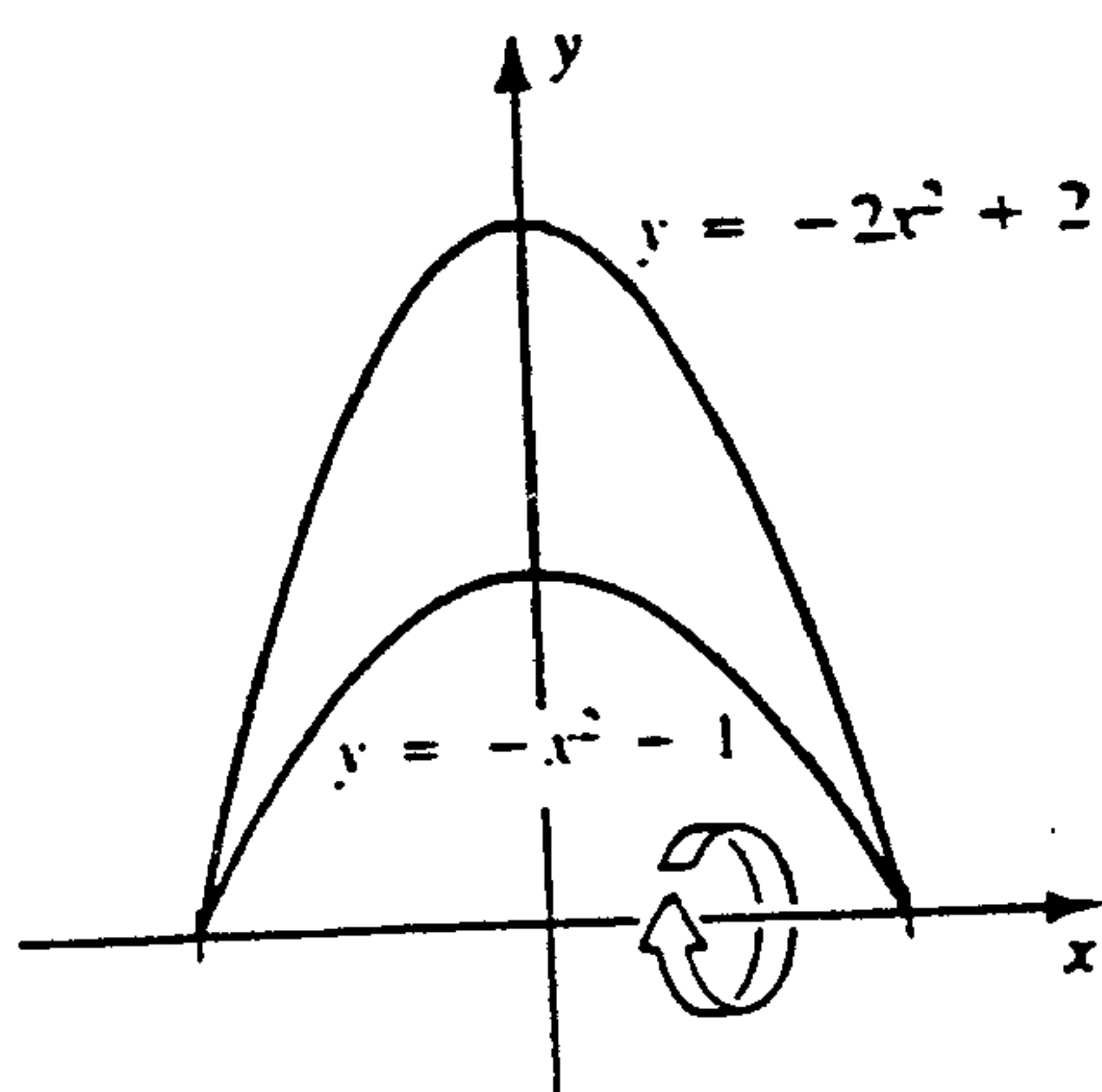
### Question 7

( Start a new page. )

- a) A body moves in a straight line. At time  $t$  seconds its acceleration is given by  $a = 6t + 1$ . At  $t = 0$  the body is at the origin and its velocity is  $-2$  m/s.

- Show that the velocity is given by  $v = 3t^2 + t - 2$ .
- Determine when the particle is at rest.
- Describe the motion.

b)



The area between the two parabolas  $y = -2x^2 + 2$  and  $y = -x^2 + 1$  is rotated about the x axis.

Find the volume of the solid thus generated.

### Question 8

( Start a new page. )

- a)
- Simplify  $\log_2 18 - 2 \log_2 \sqrt{3}$ .
  - Sketch the graph of  $y = \ln x$ .
  - Solve the equation  $2 \ln x - \ln(2x + 3) = 0$ .

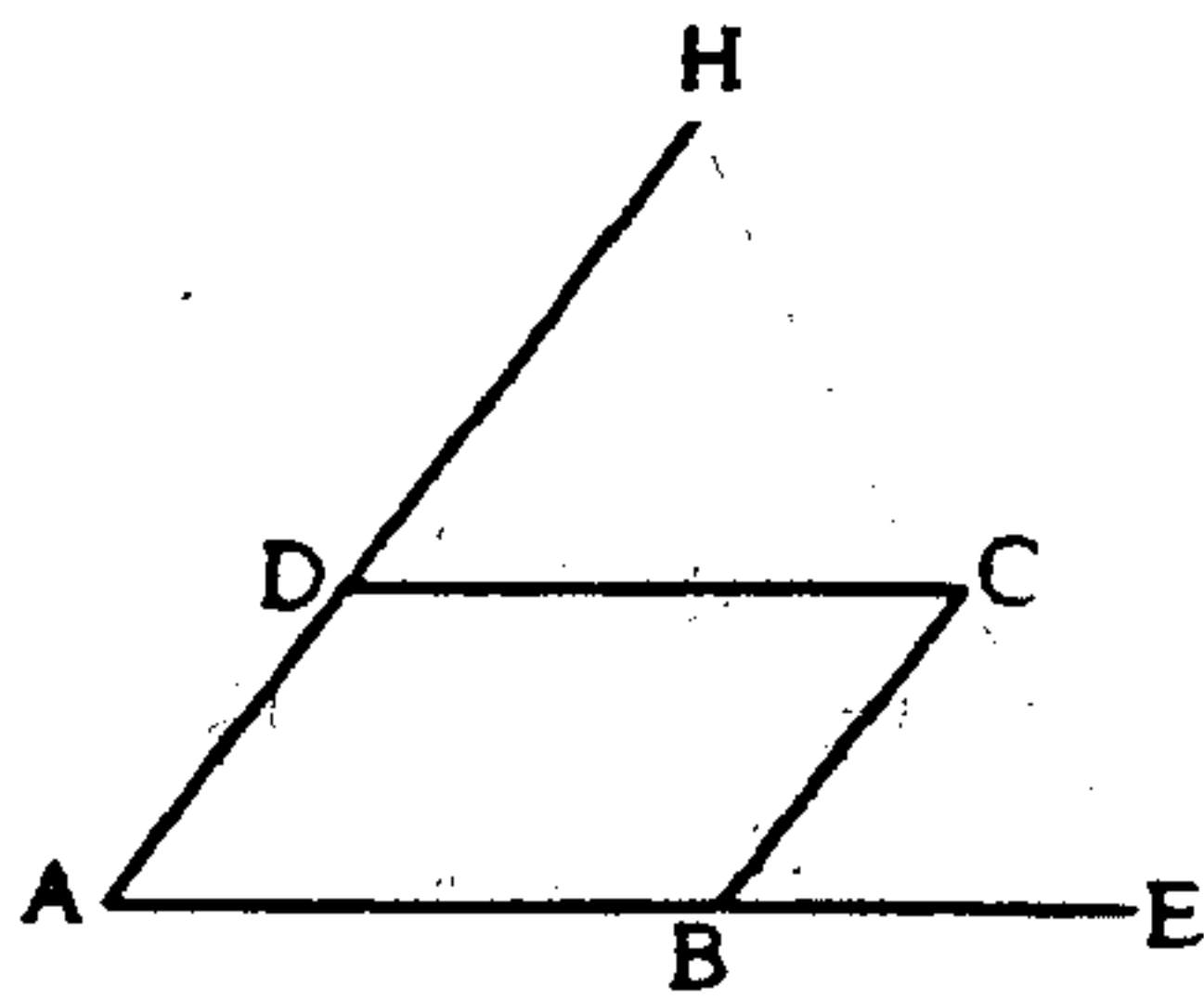
- b) A school is divided in two parts : Senior school with 400 boys and 200 girls and Junior school with 400 girls and 300 boys. A first student is chosen at random from the whole school. If this student comes from the Junior school, a second student is chosen from the Senior school; if the first student comes from the Senior school then a second student is chosen from the Junior school. By making use of a tree diagram, or otherwise, find the probability that :

- the second student chosen will be a girl;
- if the second student chosen is a boy, he is from the senior school.

**Question 9**

( Start a new page. )

- a) In a parallelogram ABCD the sides AB and AD are extended to E and H respectively so that  $\frac{AD}{DH} = \frac{BE}{AB}$ . Prove that :



- (i) the triangles HDC and CBE are similar;  
(ii) the points H, C and E lie on a straight line.

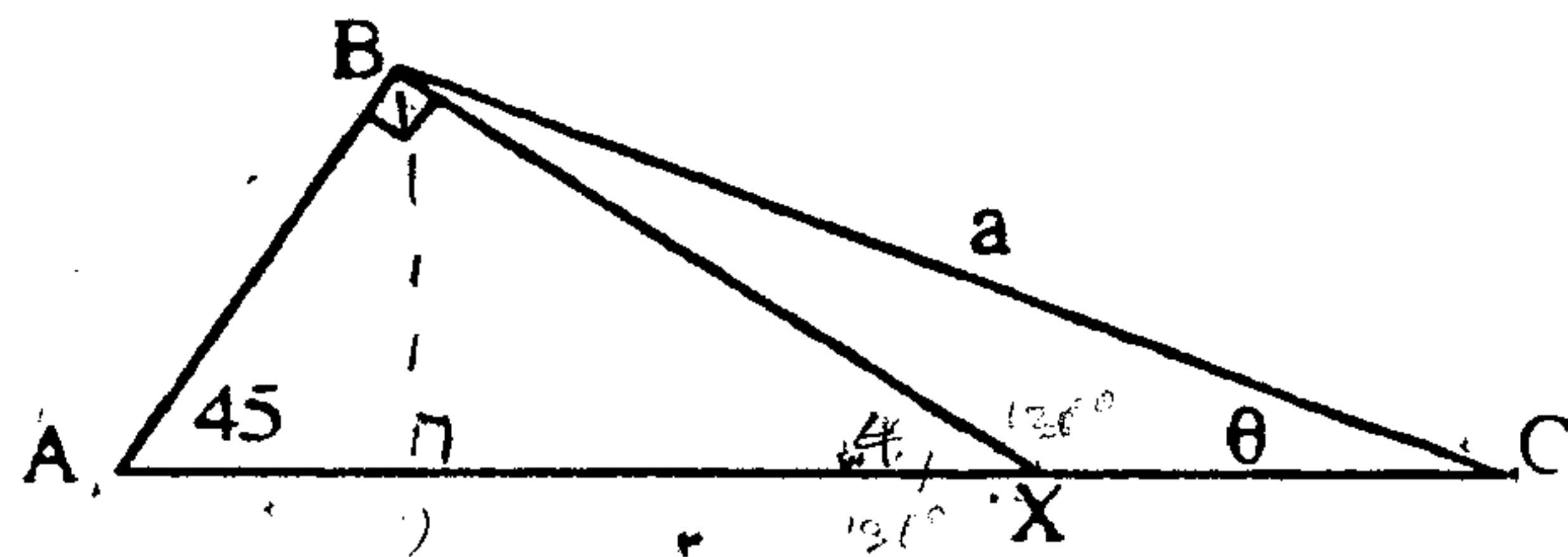
- b) A prize fund is set up with a single investment of \$20,000, to provide an annual prize of \$1,500. The fund accrues interest at 5% per annum, paid yearly. If the first prize is awarded one year after the investment, find :

- (i) the value of the prize fund immediately after the first prize has been awarded;  
(ii) the number of years for which the full prize can be awarded.

**Question 10**

( Start a new page. )

- a) (i) In  $\triangle ABC$ , BX is perpendicular to AC. Prove that  $XC = a(\cos \theta - \sin \theta)$ .  
(ii) Hence find the exact value of XC when  $a = 4$  cm and  $\theta = 30^\circ$ .



(not to scale)

- b) One thousand trout, each one a year old, are introduced into a large pond. The number still alive after  $t$  years is predicted to be

$$N = 1000 e^{-0.205t}$$

- (i) Show that the number of trout decreases at a rate proportional to the number of trout alive.  
(ii) The weight  $W(t)$  ( in kg ) of an individual trout is expected to increase according to the formula  $W(t) = 0.1 + 0.5t$ .  
After approximately how many years is the total weight (in kg) of all the trout in the pond a maximum ?

**The End**

Q1: (i)  $m \times m = -1$   
 (a)  $-3 \times \frac{1}{3} = -1 \therefore PQ \perp QR$

(ii)  $PQ = \sqrt{10}$   
 $QR = \sqrt{10} \therefore \Delta PQR$  is isosceles

(iii)  $S(4,4)$

(b)  $x = \frac{8}{3}$

(c)  $A = 9\sqrt{2}$

(d) \$11,658

Q2:

(a) (i)  $xe^{2x}(2+x)$

(ii)  $-\tan x$

(b) (i)  $1.5^\circ$

(ii)  $3\text{cm}^2$

(c) —

(d) (i)  $\sin x$

(ii)  $3x$

Q3: (i)  $\frac{(4x+3)^6}{24} + C$

(a) (ii)  $\frac{\sqrt{3}}{4}$

(b) (i) 11kms

(ii)  $59^\circ$

(c)  $x = 30^\circ, 150^\circ$

Q4: (i)  $x = 35^\circ$

(a) (ii) —

(b) (i)  $bx$

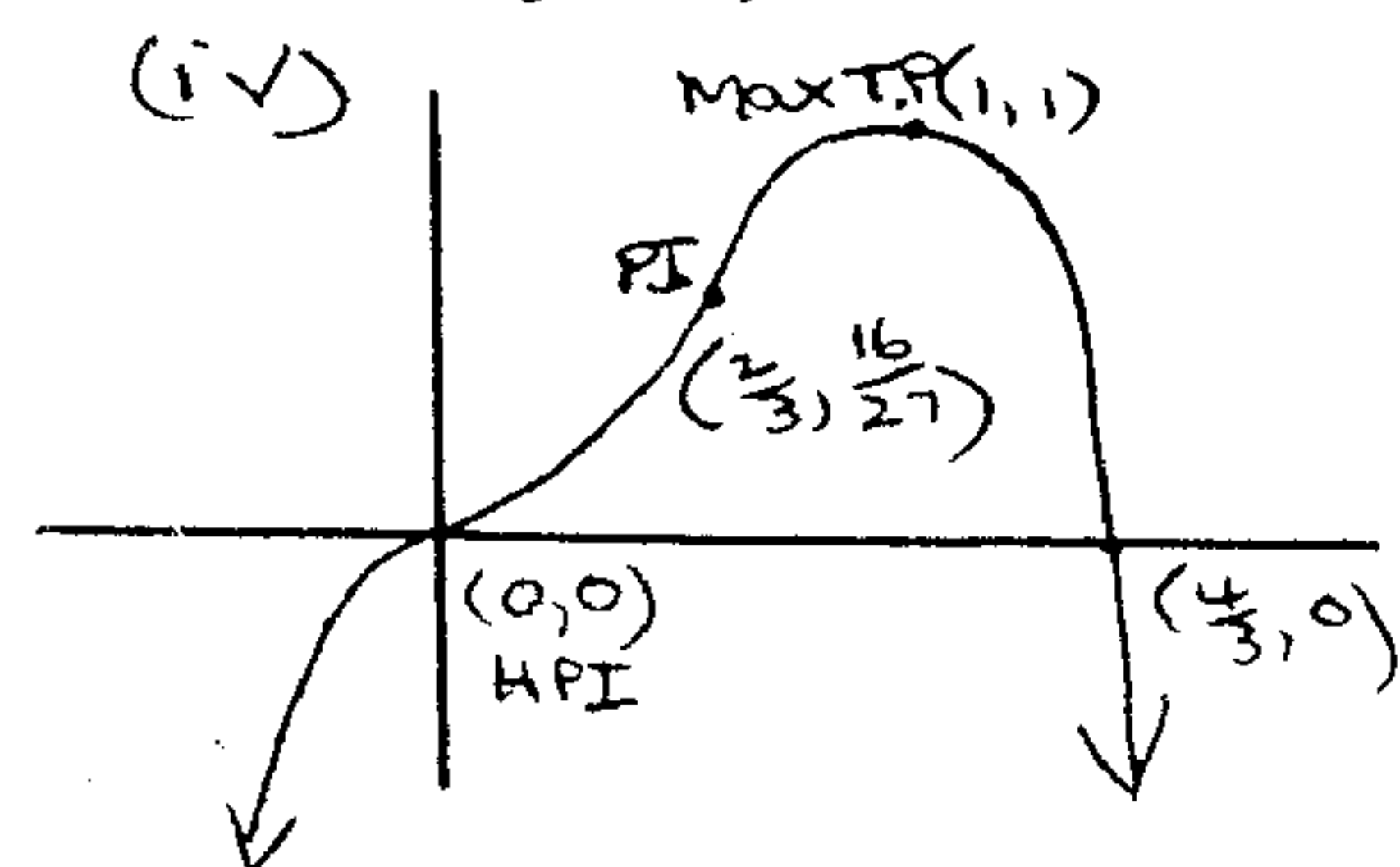
(ii)  $x < -\frac{2\sqrt{3}}{3}$

(iii)  $y = x^3 - 4x + 1$

Q5: (i)  $(0,0), (\frac{4}{3}, 0)$

(a) (ii)  $(0,0)$  Horizontal P.I.  
 $(1,1)$  Max T.P.

(iii)  $(0,0)$  already found  
 $(\frac{2}{3}, \frac{16}{27})$

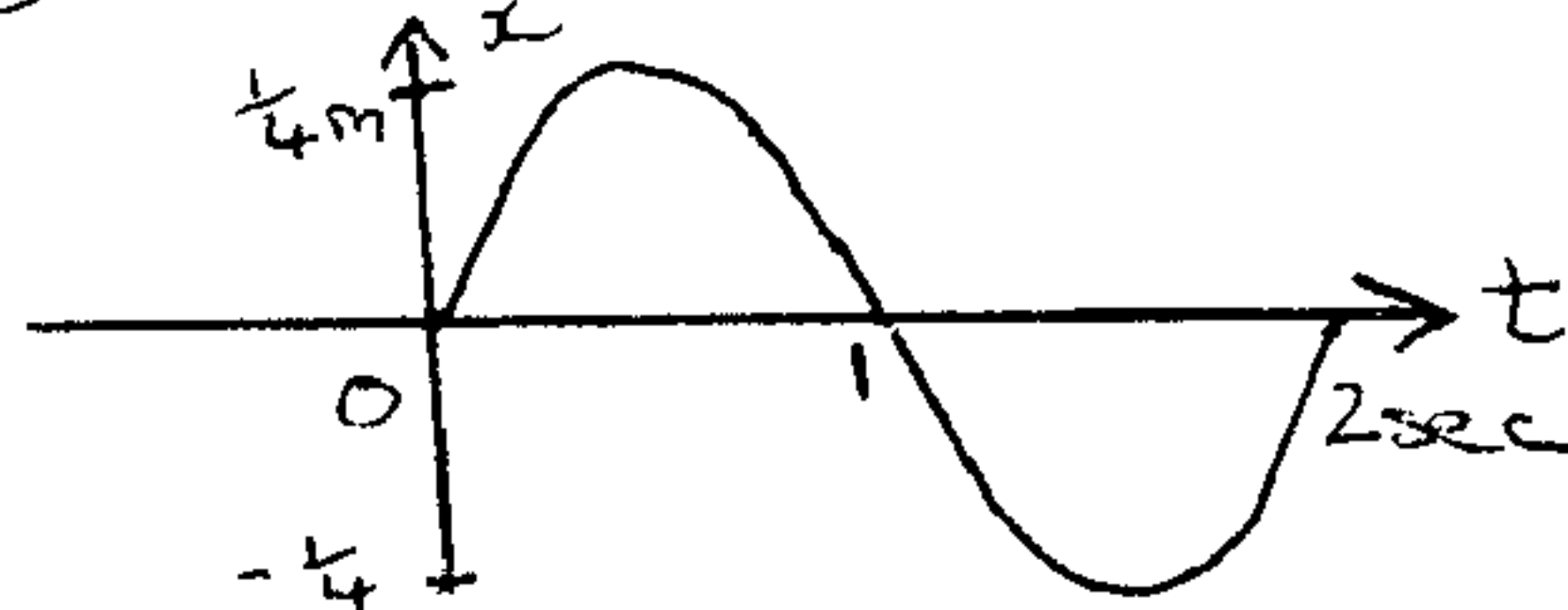


(b)  $\frac{256}{405} v^2$

Q6: (i)  $\frac{1}{6}$

(a) (ii)  $\frac{5}{36}$

(b) (i)  $x = \frac{1}{4} \sin(\pi t)$



(ii)  $2\text{km}$

(iii)  $\frac{\pi}{4} \text{ m/s}$

(c) 18 m

Q7: (i) —

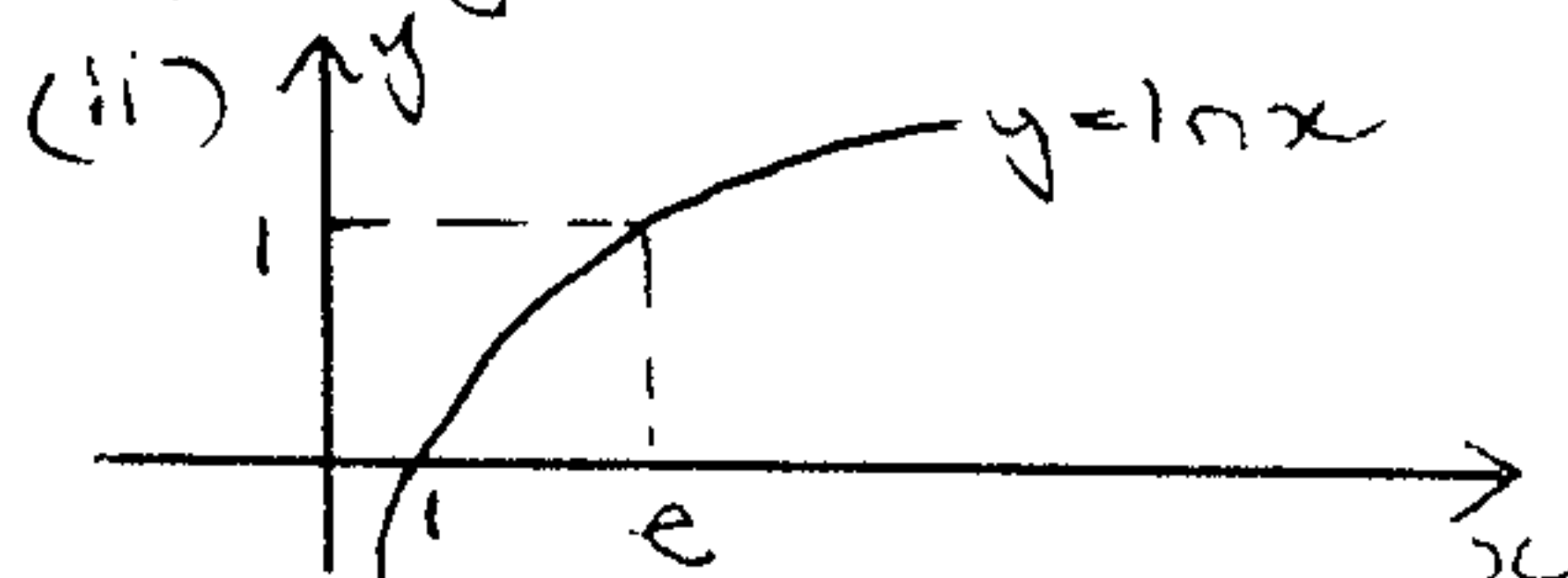
(a) (ii)  $t = \frac{2}{3} \text{ sec}$

(iii) Initially particle is at the origin and moves to the left, slowing down (since force to the right opposing motion). Particle stops at  $t = \frac{2}{3} \text{ sec}$  and then changes direction moving now towards the right.

For (b)  $V = \frac{16\pi}{5} v^3$

Q8:

(a) (i)  $\log_2 6$



(iii)  $x = 3$

(b) (i)  $\frac{121}{273}$

(ii)  $\frac{49}{76}$

Q9:

(a) (i) —

(ii) —

(b) (i) \$19,500

(ii) 22 yrs

Q10:

(a) (i) —

(ii)  $2\sqrt{3} - 2$

(b) (i)  $\frac{dN}{dt} = -0.205N$

(ii) 4.7 yrs.